Ritesh Patel

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Education

KIET Deemed to be University — Ghaziabad, India
B.Tech in Computer Science — 7.48 GPA — *Oct 2023 - June 2027*

Rani Laxmi Bai Memorial School — Lucknow, India
Class XII (ISC) — 84% — *2021-2022*

Rani Laxmi Bai Memorial School — Lucknow, India
Class X (ICSE) — 86% — *2019-2020*

Experience

Airkrit India | *Internship* | *Data Analyst* — May - Aug 2026

- Built scalable data preprocessing and analysis pipelines in Python, Pandas, and NumPy on 10K+ records, applying data cleaning, feature engineering, and anomaly detection to uncover sales and customer behavior trends.
- Created 15+ data visualizations for stakeholder decision-making and automated LLM based (GenAI) report generation, reducing manual reporting time by 30%.

Projects

RAGOps - Production RAG Evaluation & Optimization Platform | (github) — Sep 2026 - Present

- Built a configurable RAG pipeline builder supporting 4 chunking strategies, 2 embedding models, and 3 retrieval modes, enabling comparison across 20+ pipeline configurations.
- Designed and implemented a multi-index ingestion architecture that parallelizes document embeddings across registered models, reducing document processing time by 20% and allowing pipelines to switch strategies without re-processing.
- Implemented evaluation and trace-level observability across retrieval, reranking, and generation, improving retrieval accuracy by 20% while tracking Recall@K, MRR, nDCG, faithfulness, latency, token usage, and cost.

DataPilot - Ambiguity-Aware Text-to-SQL System | *FastAPI, RAG, LangChain* (github) — Aug - Sep 2026

- Built a RAG-based Text-to-SQL system using LLMs, embeddings, and vector search over PostgreSQL, achieving 87% SQL execution accuracy on 100+ test queries.
- Designed an ambiguity detection layer classifying queries as clear/ambiguous/invalid with 91% classification accuracy, reducing incorrect SQL execution by 35%.
- Built multi-turn clarification workflows for ambiguous queries (e.g., "best customer" by revenue/orders/frequency), with schema-aware retrieval across 20+ tables and 100+ columns, plus SELECT-only guardrails preventing destructive query execution.

LeaderLab - Autonomous Job Search Agent | *Next.js, LangGraph, PostgreSQL* (github) — May - July 2026

- Built a full-stack agentic job-search application with Next.js, LangGraph, and LLMs to interpret natural-language requests, autonomously run job discovery and filtering, and use LangChain tool calling for dynamic tool selection, saving 2 hours/week and reducing task latency by 10%.
- Developed resume-aware retrieval and LLM-based matching to extract candidate skills, generate relevance scores, and prioritize opportunities, increasing match relevance accuracy by 15%.
- Implemented agent state and conversational memory to retain user preferences and context across multi-step workflows, reducing redundant search inputs by 25%.

Technical Skills

AI/ML: Generative AI, LLMs, RAG, Embeddings, Vector Search, AI Agents, Model Evaluation, PyTorch, Hugging Face
Languages: Python, Java, JavaScript, SQL, Object-Oriented Programming (OOP), Data Structures and Algorithms
Frameworks: LangChain, LangGraph, Fast API, REST APIs, Next.js, React
Cloud, Databases and DevOps: AWS, PostgreSQL, SQL, Docker, Git, GitHub, CI/CD

Certifications

AWS Certified Machine Learning Engineer(MLA-C01) - Associate — May 2026

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Sep 2025